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SESSION XI

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Maximum Likelihood Estimation

ML Estimators — generally consistent
— under certain regularity conditions are most efficient as the sample size tends to infinity (asymptotic sense)
— for small sample size, estimators may be biased.

Consider the problem: (Rohatgi / Saleh)

$X \sim b(n, p)$. It is known that n is either 2 or 3 and $p = \frac{1}{2}$ or $\frac{1}{3}$. One observation is available say $X = x$.. for each possible pair (n, p):

x	$(2, \frac{1}{2})$	$(2, \frac{1}{3})$	$(3, \frac{1}{2})$	$(3, \frac{1}{3})$	Max. prob.
0	$\frac{1}{4}$	$\frac{4}{9}$	$\frac{1}{8}$	$\frac{8}{27}$	$\frac{4}{9}$
1	$\frac{1}{2}$	$\frac{4}{9}$	$\frac{3}{8}$	$\frac{12}{27}$	$\frac{1}{2}$
2	$\frac{1}{4}$	$\frac{1}{9}$	$\frac{3}{8}$	$\frac{6}{27}$	$\frac{3}{8}$
3	0	0	$\frac{1}{8}$	$\frac{1}{27}$	$\frac{1}{8}$

We find

$$(\hat{n}, \hat{p})(x) = \begin{cases} (2, \frac{1}{3}) & x = 0 \\ (2, \frac{1}{2}) & x = 1 \\ (3, \frac{1}{2}) & x = 2 \\ (3, \frac{1}{2}) & x = 3 \end{cases}$$

Maximizes the prob. of the value observed. $\Rightarrow$ More probable it came from $b(\hat{n}, \hat{p})$.

This suggests: Select an estimate θ for which the observed sample is "<u>most likely</u>" to occur.

From Rohatgi + Saleh

Principle of maximum likelihood

Chooses as the estimator that value of the parameter which maximizes the pdf (or pmf) $f(x, \theta)$. $(\theta = (\theta_1, \theta_2, \cdots \theta_k))$

Defn: Let $(X_1, X_2, \ldots, X_n)$ be a random vector with pdf (or pmf) $f_\theta(x_1, x_2, \ldots, x_n)$, $\theta \in \Theta$.
The function considered as a function of θ is called the likelihood function.

Note $x_1, x_2, \cdots x_n$ are iid.

$$L(\theta; x_1, x_2, \cdots, x_n) = \prod_{i=1}^{n} f(x_i, \theta) \qquad (1)$$

Principle of MLE

— This method consists of choosing as an estimator of θ a $\hat{\Theta}(\underset{\sim}{X})$ that maximizes $L(\theta; x_1, x_2, \cdots, x_n)$ i.e.

$$L(\hat{\theta}, x_1, x_2, \cdots, x_n) = \sup_{\theta \in \Theta} L(\theta, x_1, \cdots x_n) \qquad (2)$$

We have

$$\log L(\hat{\theta}, x_1, \cdots x_n) = \sup_{\theta \in \Theta} \log L(\theta; x_1 \cdots x_n) \qquad (3)$$

Likelihood equations.

$$\frac{\partial \log L(\hat{\theta}, x_1 \cdots x_n)}{\partial \theta_j} = 0 \qquad j = 1, \cdots n \qquad (4)$$

$$\underset{\sim}{\theta} = (\theta_1, \theta_2 \cdots \theta_k)$$

Root of Likelihood eqns yield MLE.

Example

- To estimate the probability p of a successful transmission.
- n messages transmitted and k successful transmissions.
- obtain MLE of the parameter p.

Noting that transmission of a single message can be modelled by a Bernoulli rv with the pmf

$$p_X(x) = P(X=x) = \underline{\underline{p^x (1-p)^{1-x}}}, \quad x=0,1 \quad 0 \le p \le 1 \qquad (5)$$

Joint probability of rvs $X_1, X_2, \ldots, X_n$ taking values $X_1 = x_1, X_2 = x_2, \ldots X_n = x_n$ is

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) = \prod_{i=1}^{n} p^{x_i} (1-p)^{1-x_i}$$

$$= p^{\sum x_i} (1-p)^{n - \sum x_i} \qquad (6)$$

Note that the observed values in this pmf with unknown parameter p is called a Likelihood function

$$L(p) = p^{\sum x_i} (1-p)^{n - \sum x_i} \qquad (7)$$

Maximizing $L(p)$ is equivalent to maximizing $\ln L(p)$, we have

$$\ln L(p) = \left(\sum_{i=1}^{n} x_i\right) \ln p + \left(n - \sum x_i\right) \ln (1-p)$$

Taking the derivative and setting it equal to zero, we have

$$\frac{d}{dp} \ln L(p) = \left(\sum x_i\right) \frac{1}{p} - \left(n - \sum x_i\right) \frac{1}{1-p}, \quad 0 < p < 1 \qquad (8)$$
$$= 0$$

and get

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{X} \qquad (9)$$

Verify the second derivative of $\left.\frac{d^2 L(p)}{d p^2}\right|_{p=\hat{p}} < 0$

Accordingly, $\underline{\frac{\sum x_i}{n} = \bar{X} \text{ is MLE of } p}$. (10)

<u>Note</u> $\sum x_i$ = no. of successfully transmitted messages.
The estimator of p is simply the <u>relative frequency</u>.

Discrete Population

Joint probability of the event $[X_1 = x_1, \ldots, X_n = x_n]$

Likelihood function

$$L(\theta) = P[X_1 = x_1, \ldots, X_n = x_n \mid \theta]$$
$$= \prod_{i=1}^{n} p_{X_i}(x_i \mid \theta). \qquad (11)$$

M.L.E of Θ is the solution of simultaneous equations

$$\frac{\partial L(\Theta)}{\partial \theta_i} = 0 \qquad i = 1, 2, \ldots, k \qquad (12)$$

<u>Difficulties</u>:

(i) Generally the method works fairly well.

— There may be no closed-form solution

(ii) The solution to the likelihood equations <u>may not be</u> in the parameter space.

In this case constrained maximization becomes necessary.

<u>Ex 2</u> Arrival rate of calls $X \sim P(\lambda)$, λ unknown.

Random sample $X_1 = x_1, \ldots, X_n = x_n$

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{1}{x_1! \cdots x_n!} e^{-n\lambda} \lambda^{\sum x_i} \qquad (13)$$

or $\ln L(\lambda) = -\ln(x_1! \cdots x_n!) - n\lambda + (\sum x_i) \ln \lambda$

Taking derivative with respect to λ, we get

$$-n + \frac{\sum x_i}{\lambda} = 0$$

$$\text{or } \hat{\lambda} = \frac{1}{n} \sum x_i = \bar{X} \qquad (14)$$

Ex 3

X = "Time to failure of a telephone switching system"

$X \sim Exp(\lambda)$

A random sample $X_1 = x_1, \dots, X_n = x_n$ of n times to failure

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i} \quad (15)$$

$$\frac{dL(\lambda)}{d\lambda} = n\lambda^{n-1} e^{-\lambda \sum x_i} - (\sum x_i)\lambda^n e^{-\lambda \sum x_i} = 0$$

<u>MLE estimator of the failure rate</u>

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\bar{X}} \quad (16)$$

MLE estimator of the failure rate to be the reciprocal of the sample mean.

<u>Ex 4</u>. Let $X_1, X_2, \dots, X_n$ be a sample from $U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$. Obtain MLE of θ.

<u>Soln</u>

$$L(\theta, x_1, x_2, \dots, x_n) = \begin{cases} 1 & \text{if } (\theta - \frac{1}{2}) \le \min(x_1, \dots x_n) \\ & \le \max(x_1, \dots x_n) \le \theta + \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

$L(\theta, \underset{\sim}{x})$ attains max. when

$$\theta - \frac{1}{2} \le \min(x_1, \ldots x_n) \text{ and } \left(\theta + \frac{1}{2}\right) \geq \max(x_1, \ldots x_n) \quad (17)$$

or $\boxed{\theta < \min(x_1, \ldots x_n) + \frac{1}{2} \text{ and } \theta \geq \max(x_1, \ldots x_n) - \frac{1}{2}}$

Accordingly, <u>every</u> statistic such that

$$\max_{1 \le i \le n} X_i - \frac{1}{2} \le T(X_1, \ldots X_n) \le \min_{1 \le i \le n} X_i + \frac{1}{2} \quad (18)$$

is an MLE of θ

Therefore we can construct MLE statistic of θ,

$$T_\alpha(X_1, \ldots X_n) = \max_{1 \le i \le n} X_i - \frac{1}{2} + \alpha\left(1 + \min_{1 \le i \le n} X_i - \max_{1 \le i \le n} X_i\right) \quad (19)$$

for $0 \le \alpha < 1$.

$T_\alpha(X_1, \ldots X_n)$ is an MLE of θ.

What is $T_{1/2}(X_1 \ldots X_n)$?

Put $\alpha = \frac{1}{2}$ in (19)

$$T_{1/2}(X_1, \ldots X_n) = \frac{\min X_i + \max X_i}{2} \quad (20)$$

is an MLE of θ

(From Rohatgi/Saleh)

Example

Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution with pdf,

$$f(x|\theta) = \begin{cases} \frac{1}{\theta} & 0 < x < \infty, \ \theta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Obtain MLE of θ.

Soln

$$L(\theta) = \prod_{i=1}^{n} f(x_i|\theta) = \left(\frac{1}{\theta}\right)^n \qquad (21)$$

Likelihood eqn

$$\ln L(\theta) = -n \ln \theta \qquad (22)$$

This yields absurd result.

How to obtain MLE in this case:

Arrange n observations such that

$$0 \leq x_{(1)} \leq x_{(2)} \cdots \leq x_{(n)} \leq \theta \qquad (23)$$

The minimum value of θ consistent with the sample is $x_{(n)}$

Hence $\hat{\theta} = x_{(n)}$ (24)